EXERCISE 02 - SQL FOUNDATIONS

SQL AGGREGATE FUNCTIONS & OPERATORS

QUESTION 1

```
SELECT DISTINCT department
FROM students;
```

output :

department
IT
HR
Finance

QUESTION 2

```
SELECT department,
       ~~AGG~~ AVG(age) AS avg_age
FROM students
GROUP BY department;
```

output :

department	avg-age
IT	20.5
HR	22
Finance	23

QUESTION 3

```
SELECT department,
       COUNT(*) AS student_count
FROM students
GROUP BY department
HAVING COUNT(*) > 1;
```

output :

department	student_count
IT	2
HR	2

QUESTION 4

```
SELECT *
FROM   students
WHERE  age BETWEEN 21 AND 23;
```

output :

student_id	name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

QUESTION 5

```
SELECT *
FROM   students
WHERE  department IN ('IT', 'HR')
       AND age > 21;
```

Output :

student_id	name	age	department
2	Bob	22	HR
3	Eve	22	HR

02 THE COURSES TABLE

QUESTION 6

```
SELECT department,
    sum(credits) AS total_credits
FROM courses
GROUP BY department
HAVING sum(credits)>5;
```

output:

department	total_credits
IT	11

QUESTION 7

```
SELECT *
FROM courses
WHERE NOT credits =4;
```

output:

Course_id	Course_name	department	Credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

QUESTION 8

```
SELECT course_id, course_name, credits
FROM courses
ORDER BY credits DESC
LIMIT 3;
```

output:

Course_id	Course_name	Credits
102	Python	4
103	Data Science	4
101	SQL Basics	3

03 THE ENROLLMENTS TABLE

QUESTION 9

```
SELECT  MAX(grade) AS max_grade,
        MIN(grade) AS min_grade,
        AVG(grade) AS avg_grade
FROM    enrollments;
```

output:

Max_grade	Min_grade	avg_grade
90	78	84.6

QUESTION 10

```
SELECT  course_id,
        COUNT(enrollment_id) AS enrollment_count
FROM    enrollments
GROUP BY course_id;
```

output:

Course_id	enrollment_count
101	1
102	1
103	1
104	1
105	1

Q4 THE SALARIES TABLE

QUESTION 11

```
SELECT department,
       SUM(salary) AS total_salary,
       SUM(bonus) AS total_bonus
FROM salaries
GROUP BY department;
```

output:

department	total_salary	total_bonus
IT	122 000	10500
HR	109 000	7500
Finance	70 000	6000

QUESTION 12

```
SELECT department, AVG(salary) AS avg_salary
FROM salaries
GROUP BY department
HAVING AVG(salary) > 55000;
```

output:

department	avg_salary
IT	61 000
Finance	70 000

QUESTION 13

```
SELECT employee_id, name, salary, bonus,
       (salary + bonus) AS total_compensation
FROM Salaries
WHERE (salary+bonus) > 60 000;
```

output:

employee_id	name	Salary	bonus	total_compensation
1	Tom	60000	5000	65000
3	Spike	70000	6000	76000
4	Tyke	62000	5500	67500

OF THE PROJECTS TABLE

QUESTION 14

```
SELECT department, SUM (budget) AS total_budget,
       AVG(budget) AS avg_budget
FROM projects
GROUP BY department
HAVING AVG (budget) > 70000;
```

output:

department	total_budget	avg_budget
IT	270000	135000
Finance	80000	80000

QUESTION 15

```
SELECT *
FROM projects
WHERE budget BETWEEN 50000 AND 120000
AND NOT department = 'Marketing';
```

output:

Project_id	project_name	department	budget
1	AI App	IT	120000
2	Payroll System	Finance	80000
5	HR portal	HR	50000